Lecture 29

Features of Turbulent Flows and Approaches to Their Simulation

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Contents

- Properties of turbulent flows.
- Laminar-turbulent transition.
- Coherent structures.
- Turbulent flows in engineering.
- Approaches to studying turbulence.

Properties of Turbulent Flows, I

- **Unsteadiness:** Apparently random variation in time of all variables at any point in a turbulent flow (but not quite chaotic...). The time- or ensemble-averaged fields are smooth and may be steady.

- **Three-dimensionality:** Except for some special situations, turbulent flows are always three-dimensional; only the time-averaged field can be two- or even one-dimensional...

- **Vorticity:** Turbulent flows contain a high level of vorticity; vortex stretching is one of the primary mechanisms by which the turbulence intensity is increased.

- **Stirring:** Fluid parcels from different locations are brought into contact by turbulent transport. This process is often called *turbulent diffusion,* although the transport mechanism is convection. Mixing takes place eventually by molecular diffusion; the process is much more intense than in a laminar flow, because turbulent stirring increases local gradients.

Properties of Turbulent Flows, II

- **Dissipation:** The kinetic energy of the flow is reduced since mixing is a dissipative process; some energy is irreversibly converted into internal energy of the flow (i.e. heat).

- **Coherent Structures:** Repeatable and essentially deterministic events responsible for a large part of mixing, but different from each other in size, strength and interval between occurrences...

- **Broad range of length and time scales:** Turbulent fluctuations span several orders of magnitude in both time and length scales, making *direct numerical simulation* (DNS), in which all scales are resolved, very difficult (requiring extreme computing resources).

- **Reynolds-number dependence:** With increasing Reynolds number, the range of scales rapidly increases, thus limiting DNS to a lower range of Reynolds numbers.

Laminar-Turbulent Transition, I

- Boundary layers cannot be maintained *laminar* over a long distance…
- There are three ways of transition in a boundary layer from laminar to turbulent:
  - **Natural transition:** Instabilities in the boundary layer become amplified and develop to turbulence spots that grow further downstream, leading to a fully turbulent flow.
  - **Bypass transition:** Boundary layer is initially laminar but when the free stream is turbulent, disturbances propagate from outside into boundary layer and lead to it becoming turbulent.
  - **Separation-induced transition:** Laminar separation of a boundary layer caused by an adverse pressure gradient is followed by turbulent reattachment (shear-layer instability leads to turbulence which grows into the boundary layer)...
- Natural transition is the most difficult to predict...
- Transition is sensitive to wall roughness, sound effects, free-stream fluctuations etc...

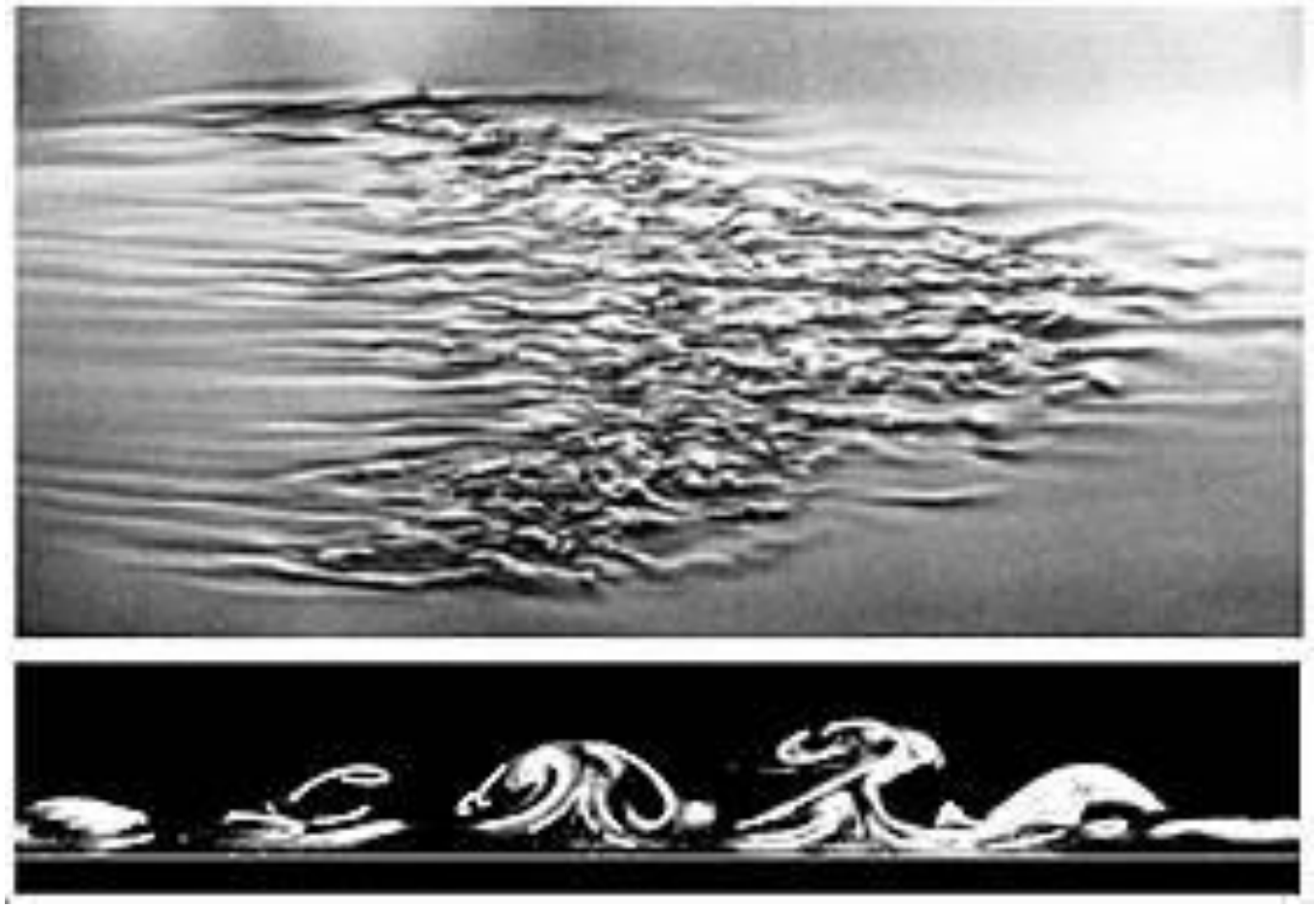
Laminar-Turbulent Transition, II

- Turbulence spots in a transitional boundary layer flow: they first appear and disappear again, but when they re-appear further downstream, the flow stays turbulent...

Visualization in an experiment

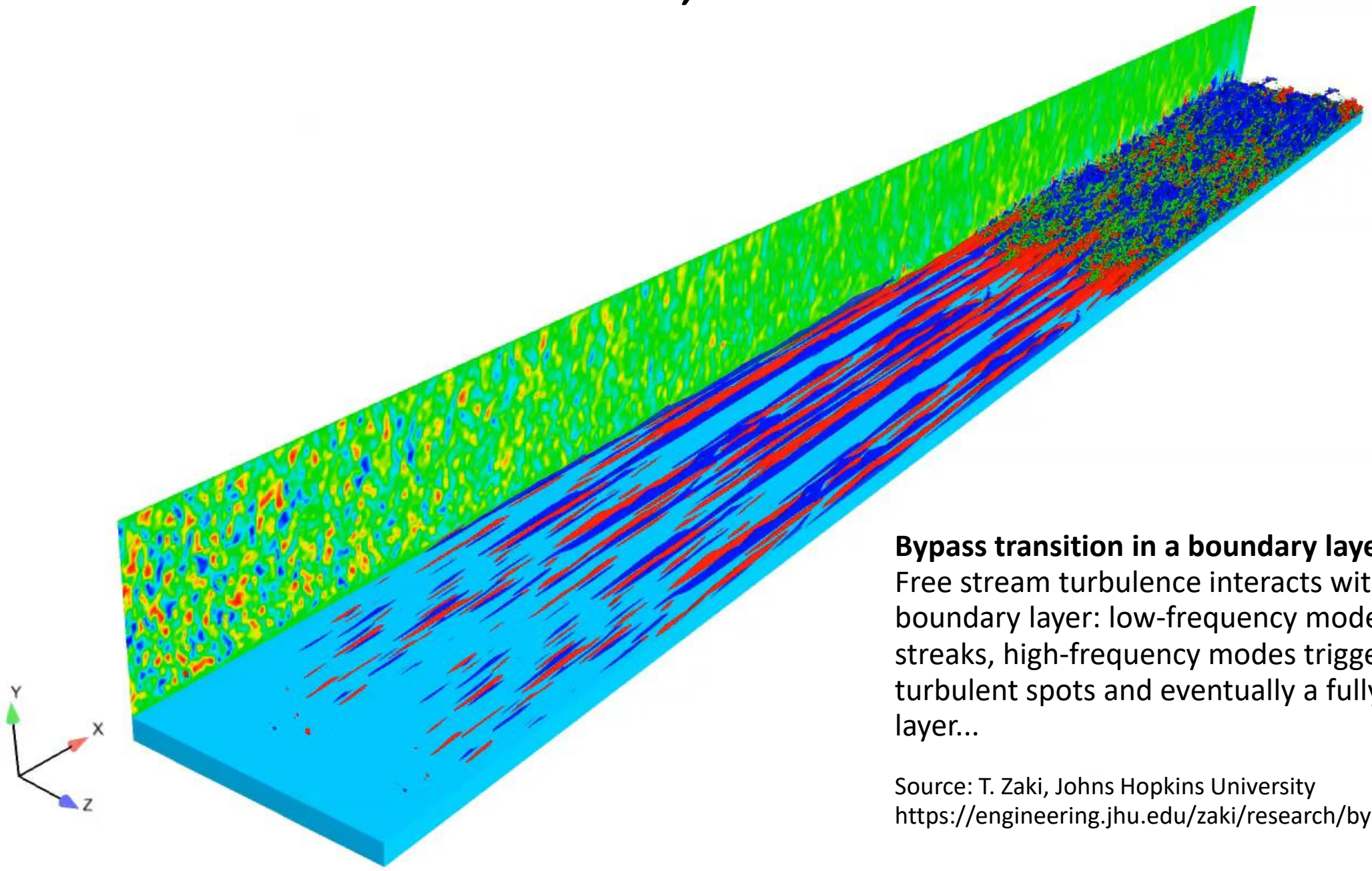
Top view

Side view



Source:
M. Van Dyke, An Album of Fluid Motion, Parabolic Press, 1986.

Laminar-Turbulent Transition, III



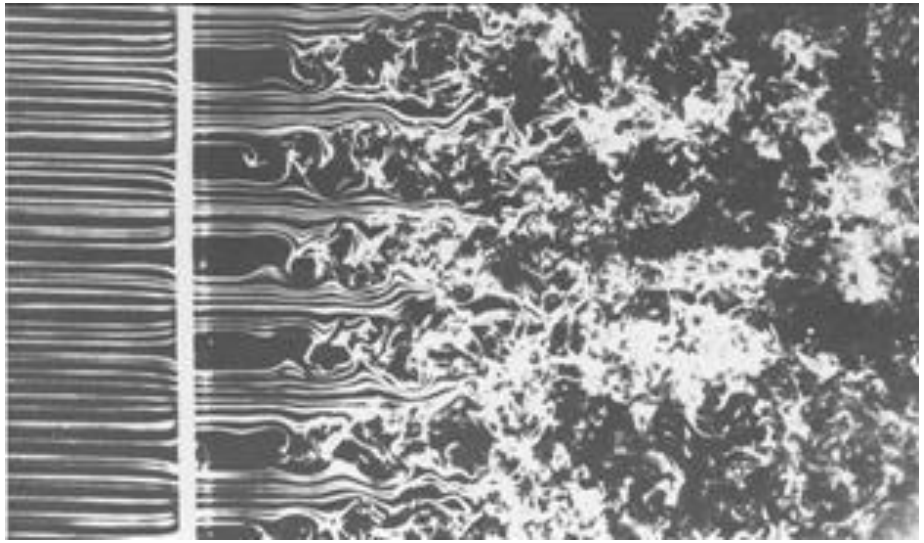
Bypass transition in a boundary layer (DNS):
Free stream turbulence interacts with an initially laminar boundary layer: low-frequency modes lead to creation of streaks, high-frequency modes trigger instability, leading to turbulent spots and eventually a fully turbulent boundary layer...

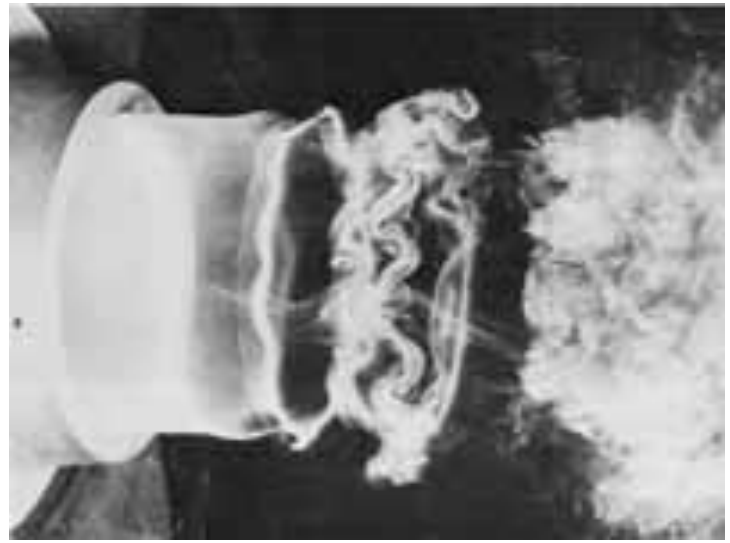
Source: T. Zaki, Johns Hopkins University
https://engineering.jhu.edu/zaki/research/bypass-transition/

Laminar-Turbulent Transition, IV

Shear layers (jets, wakes...) become turbulent because of a growing Kelvin-Helmholtz instability. Figures below show transition to turbulent flow regime behind a grid (left and bottom) and in a round jet (right).

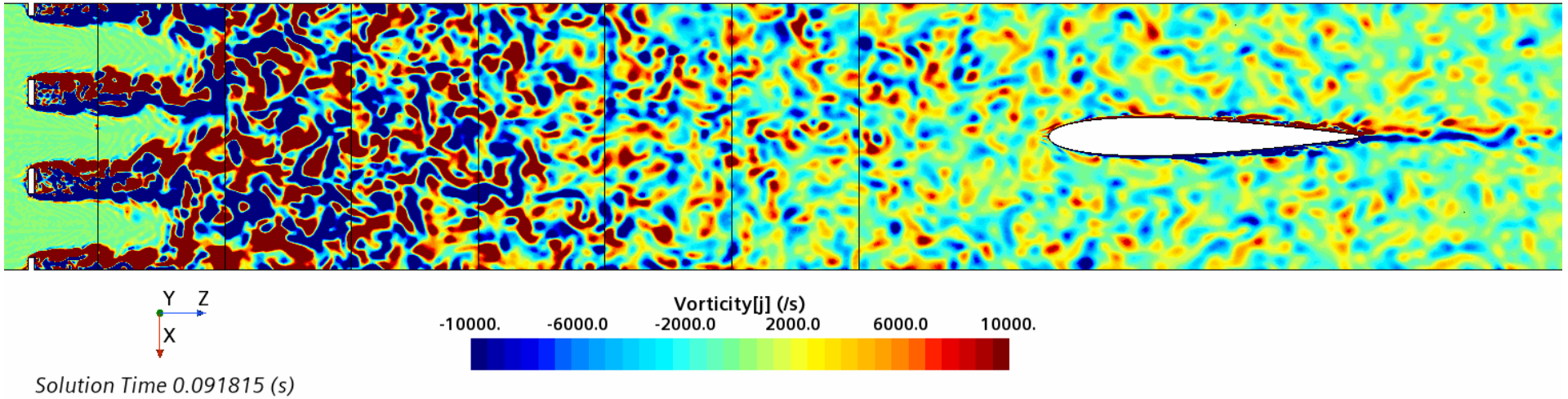
Source for photographs: M. Van Dyke, An Album of Fluid Motion, Parabolic Press, 1986.





Large-eddy simulation of turbulence generation by a grid, its spatial decay and interaction with an airfoil.

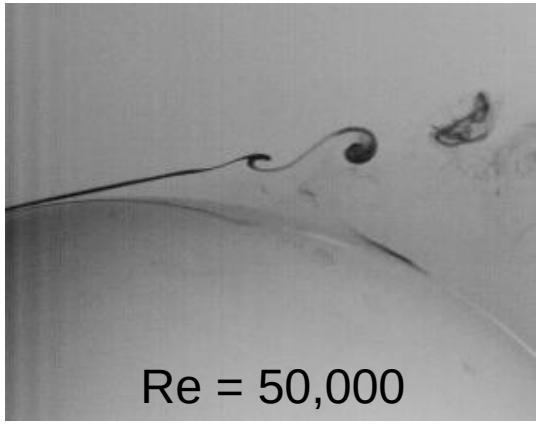
Simulation by M. Perić using *Simcenter STAR-CCM+* software.

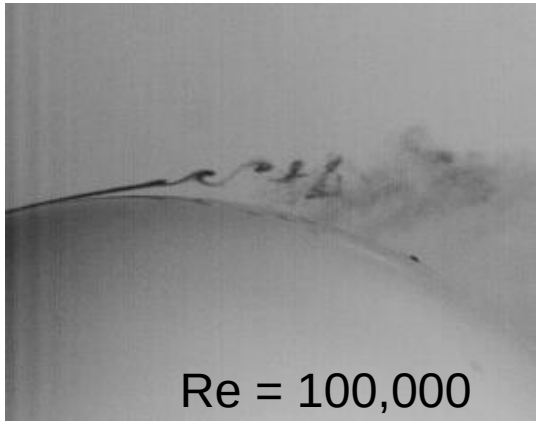


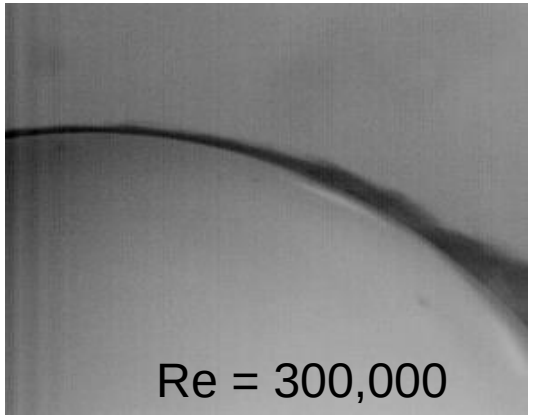
Laminar-Turbulent Transition, V

Transition to turbulence in the shear layer after laminar flow separation from a sphere:
Same pattern at each Reynolds-number, but structures become smaller with increasing Reynolds-number, until turbulent shear layer after a laminar separation re-attaches to wall around Re = 300,000 leading to *drag crises* and a delayed turbulent separation.





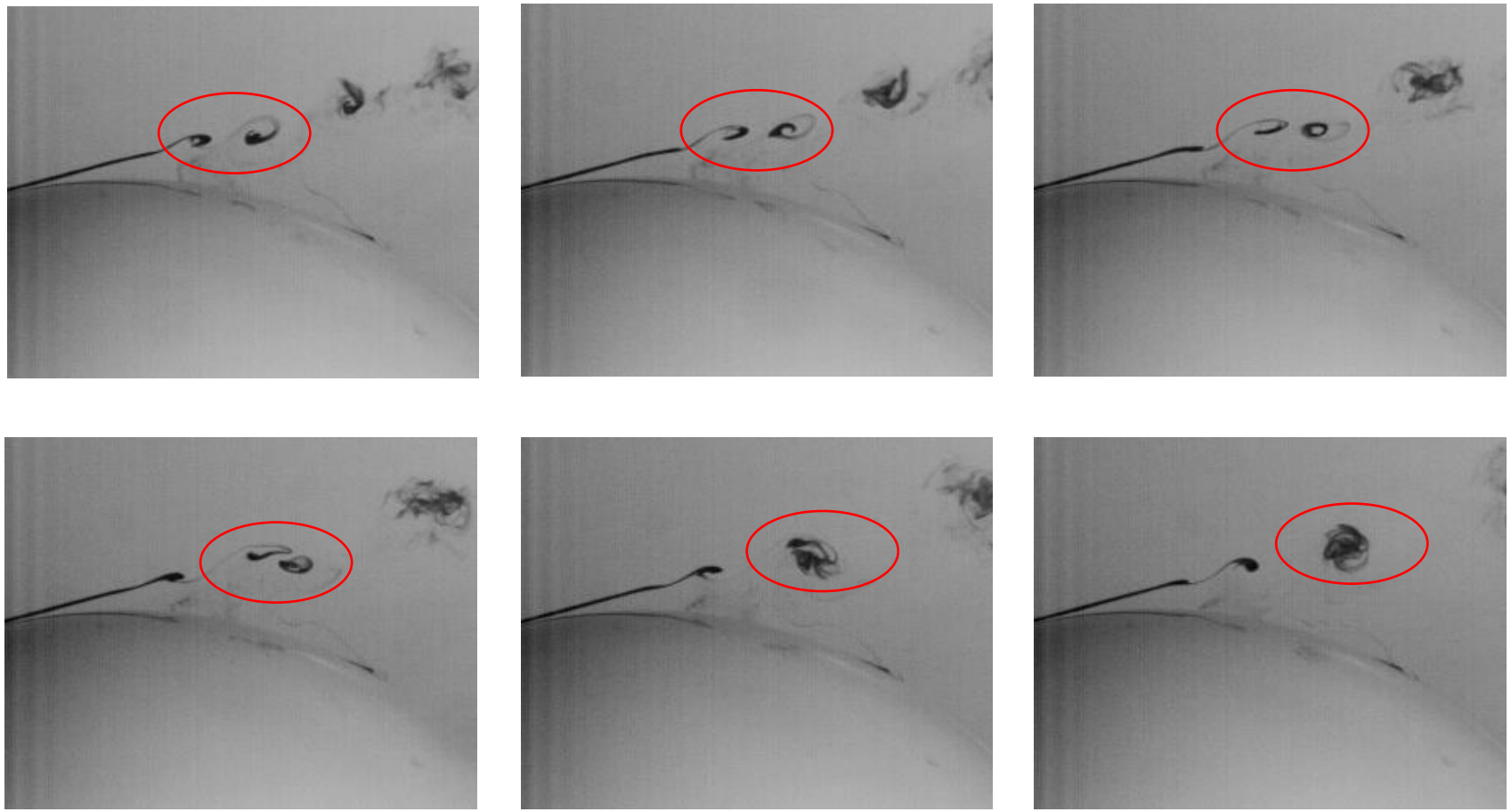




Source: Records from the towing tank of the former Institute of Shipbuilding in Hamburg, 1996.

Laminar-Turbulent Transition, VI

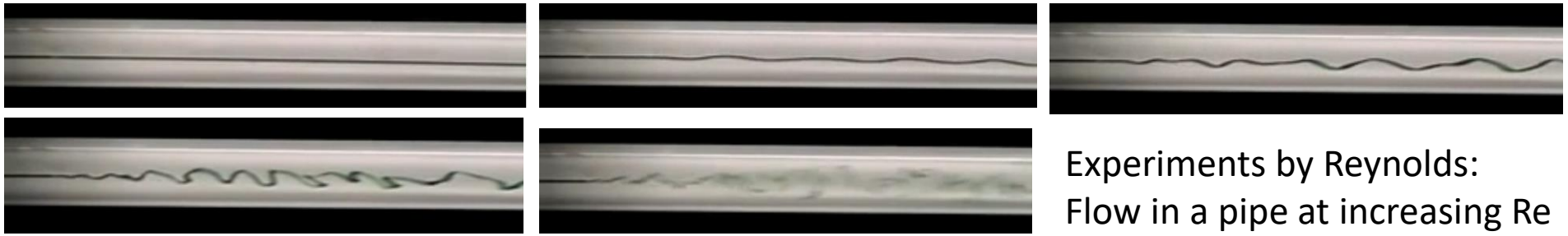
This sequence of pictures (1/500th of a second apart) shows the phenomenon of vortex pairing, creating a coherent structure that disturbs the semi-regular pattern of Kelvin-Helmholtz instability (Re = 50,000).



Source: V. Bakić, PhD theses, TUHH, 1998.

Laminar-Turbulent Transition, VII

- Flows in pipes and channels stay always laminar for Reynolds numbers below approx. 2,000 (if disturbed, the flow may become turbulent, but it returns to a laminar state after disturbances die out)...
- Flow can stay laminar even for much higher Reynolds numbers if disturbances are kept away (experiments with laminar flows in a pipe at Reynolds numbers up to 100,000 have been conducted in a controlled environment)...
- At Reynolds numbers higher than critical, any disturbance can trigger transition to turbulence – and the flow then stays turbulent even if disturbances go away...
- *Direct numerical simulation (by solving the Navier-Stokes equations using sufficiently fine grid and time steps) produces a turbulent flow at Reynolds numbers above critical – numerical solution contains sufficient disturbances through* ***round-off*** *and* ***iteration*** *errors...*



Experiments by Reynolds:
Flow in a pipe at increasing Re

Coherent Structures

- Coherent structures are partly deterministic...
- They account for a relatively small amount of energy...
- … but for a large portion of transport in a turbulent flow.
- Vorticity is strongest in coherent structures...
- They are mostly due to mean strain in the flow...
- They are different in each flow or region of it (problem-dependent and hence difficult to model)...
- They are changing all the time: being created, evolving and being destroyed...
- They can be seen as vortexes (as in slide 11).

Turbulent Flows in Engineering, I

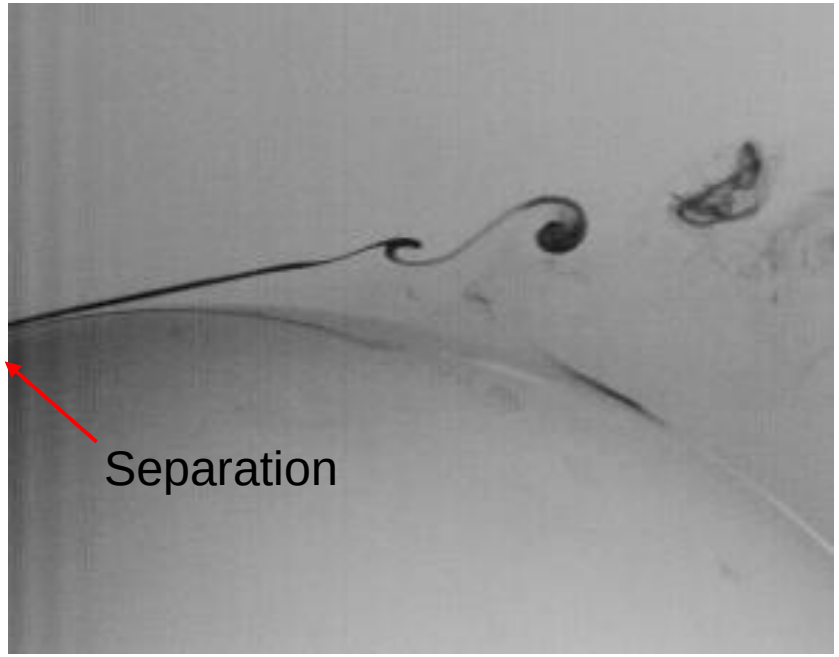
- Sometimes turbulence is *desired* in the flow:
  - When **good mixing** is needed (e.g., in mixing vessels):
    - Turbulence stirs the fluid and makes mixing far more intensive than what otherwise happens by molecular diffusion;
    - Important in chemical and process engineering, e.g., to mix fuel and oxidant or other reactants...
  - When **good heat transfer** is needed (e.g., in heat exchangers):
    - Turbulence enhances exchange of heat between fluid and walls by bringing more high-speed fluid more often closer to walls...
    - At the wall itself, heat exchange still takes place by conduction, but the temperature gradient directly at wall is much higher in a turbulent than in a laminar flow, hence leading to a higher heat-transfer rate…
    - Turbulence is important in heat exchangers of almost all kinds...

Turbulent Flows in Engineering, II

- Turbulence is also *desired* when separation should be *delayed*:
  - Turbulence brings high-momentum fluid from outside layers closer to wall, so boundary layer can longer flow against adverse pressure gradient...
  - The consequence is a delayed separation and a lower drag (*drag crises* for sphere and circular cylinder: boundary layer becomes turbulent before separation)...
  - Turbulence is thus often promoted:
    - By using surface roughness or otherwise structured surface (e.g., golf ball);
    - By boundary-layer tripping (roughness, trip wire), e.g., when doing experiments in model scale, to achieve similarity with full scale flow features;
    - By specially designed obstacles (turbulators; e.g., at airplane wings, designed to keep the boundary layer turbulent and prevent stall...).

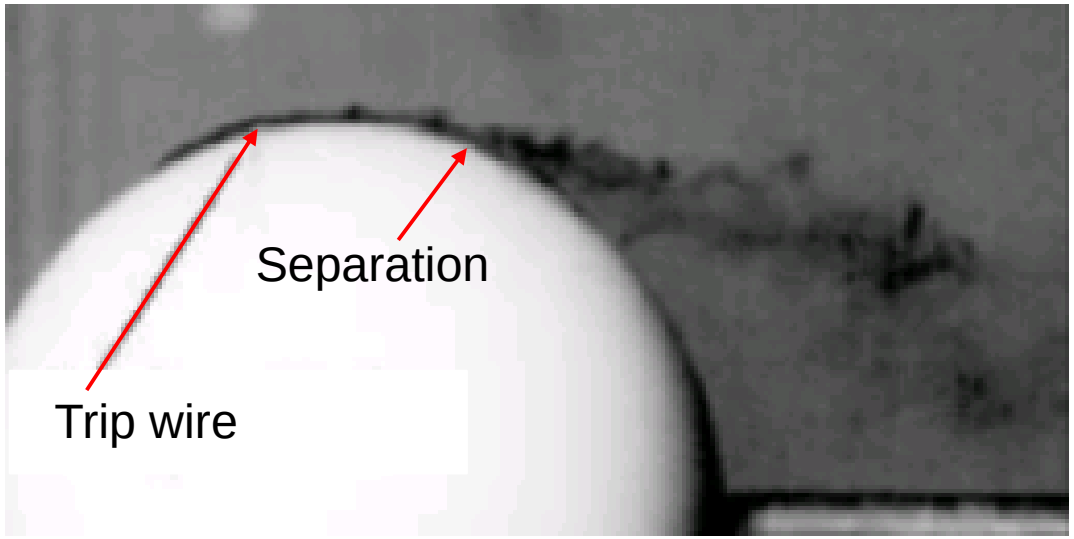
Turbulent Flows in Engineering, III

Flow around sphere at Re = 50,000



Smooth sphere: Laminar separation takes place at about 85° from front stagnation point. The wake is turbulent and the mean recirculation zone behind sphere is about 1.5 D long.

Experiments performed in a towing tank in Hamburg; sphere diameter 61.4 mm (billiard ball)

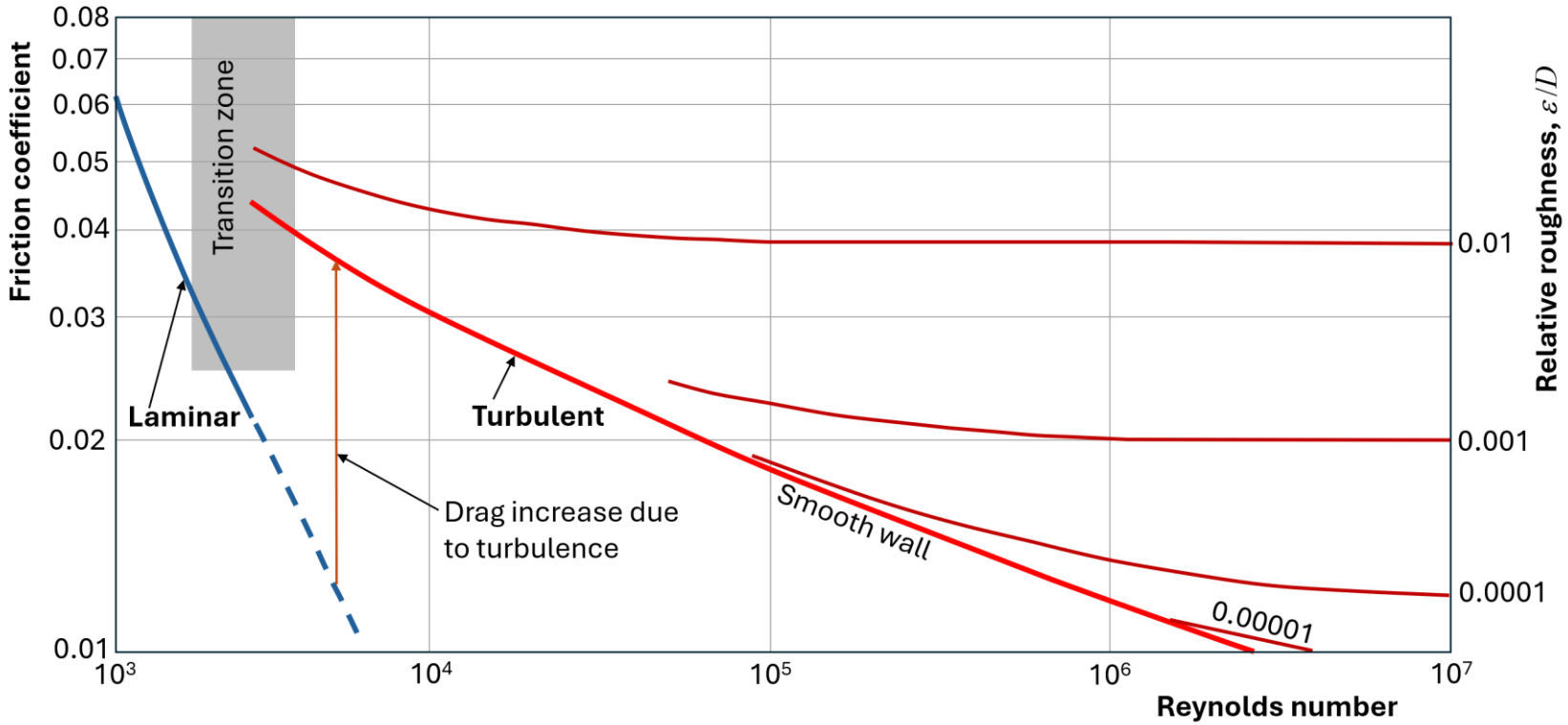


Sphere with a trip wire at 75° from front stagnation point: Boundary layer becomes turbulent downstream of trip wire. The turbulent separation takes place at about 115° from front stagnation point, the recirculation zone in the wake is about 1.0 D long, and the drag is reduced by a factor 3 compared to smooth sphere!

Sphere diameter 61.4 mm, trip wire diameter 0.5 mm

Turbulent Flows in Engineering, IV

- Often turbulence is *not desired* in the flow:
  - It increases skin friction (resulting in higher drag);
  - It increases pressure losses in pipe and channel networks (requiring more pumping power);
  - It increases *noise* (turbulence-induced noise sources) and vibration...
- Compromises have to be made in the optimization process, with attempts to control turbulence…



Friction coefficient in pipe flows

Examples of Turbulence Control



A classical jet engine has a smooth, circular edge at exhaust (left aircraft). Newer engines have so-called “chevron-nozzles” – the perimeter is not smoothly circular but is “corrugated”. The aim is to produce fine-scale turbulence at the jet edge, in order to stabilize it and reduce both vibrations and noise.



Dimples on a golf ball lead to an earlier transition of boundary layer to a turbulent state, which results in lowering of drag – hence the ball flies further with the same kick than if it was smooth…



Turbulence generators on aircraft wing, designed to increase mixing and delay separation at high angles of attack.

Approaches to Studying Turbulence, I

- **Empirical correlations:** Diagrams and equations giving relation between drag, skin friction, or heat transfer coefficient as a function of dimensionless parameters (Reynolds-number, Prandtl-number, Rayleigh-number etc.).
- An example: Blasius formula for friction coefficient in a pipe flow: $f = 0.3164\ \mathrm{Re}^{-0.25}$
- These correlations work well for simple geometries (plate; channel; pipe…) but are not helpful for flows in complex geometries.
- **Measurement techniques:**
  - Flow visualization (injection of die or bubbles; tufts on walls; oil film on walls, etc.);
  - Local measurements (pressure, velocity, temperature; Pitot-tube, Laser-Doppler-Anemometer, Hot-Wire-Anemometer...);
  - Field measurements (e.g., Particle-Image-Velocimetry)
- Measurement is time-consuming and often limited to laboratory conditions…

Approaches to Studying Turbulence, II

- **Numerical simulation:**
  - *Direct numerical simulation (DNS):* requires grid and time step that resolve all spatial and temporal variation of velocity and pressure. Requirement for computing resources increases over-proportionally with Re (approx. $\mathrm{Re}^{2.8}$).
  - DNS is mostly used to generate detailed data instead of experiments and used to validate turbulence models.
  - *Large-eddy simulation (LES):* requires grid and time step fine enough to resolve ca. 70% of turbulence (larger length and time scales), while modeling the effects of smaller (subgrid) scales.
  - These days LES is used to simulate complex engineering flows, but only up to moderately high Reynolds numbers – tendency increasing, with increasing computing power.
  - *Solution of Reynolds-averaged Navier-Stokes equations:* (eddy viscosity, algebraic, and differential Reynolds-stress models). This is the most widely used approach in industry.

Summary

- Turbulence is one of the greatest challenges in physics…
- It is present in most flows of practical interest, both in nature and in engineering…
- Sometimes is turbulence desired, sometimes not…
- Transition from laminar to turbulent flow is important; engineers often try to delay or promote it…
- Accurate simulation of turbulence is possible – it is called “Direct Numerical Simulation” – but it is not practical for most engineering flows.
- Engineers need to use turbulence models, which are not perfect. It is important to assess modeling errors, which requires knowing other errors (iteration, discretization)…
- More details on the three approaches to simulate turbulence will be given in the next three lectures.

The PowerPoint presentation file is available at the following link within Siemens Community environment:

https://community.sw.siemens.com/s/topic/0TOVb0000000GETOA2/the-peri%C4%87-lectures-on-cfd